Project Design Phase-II
Data Flow Diagram & User Stories

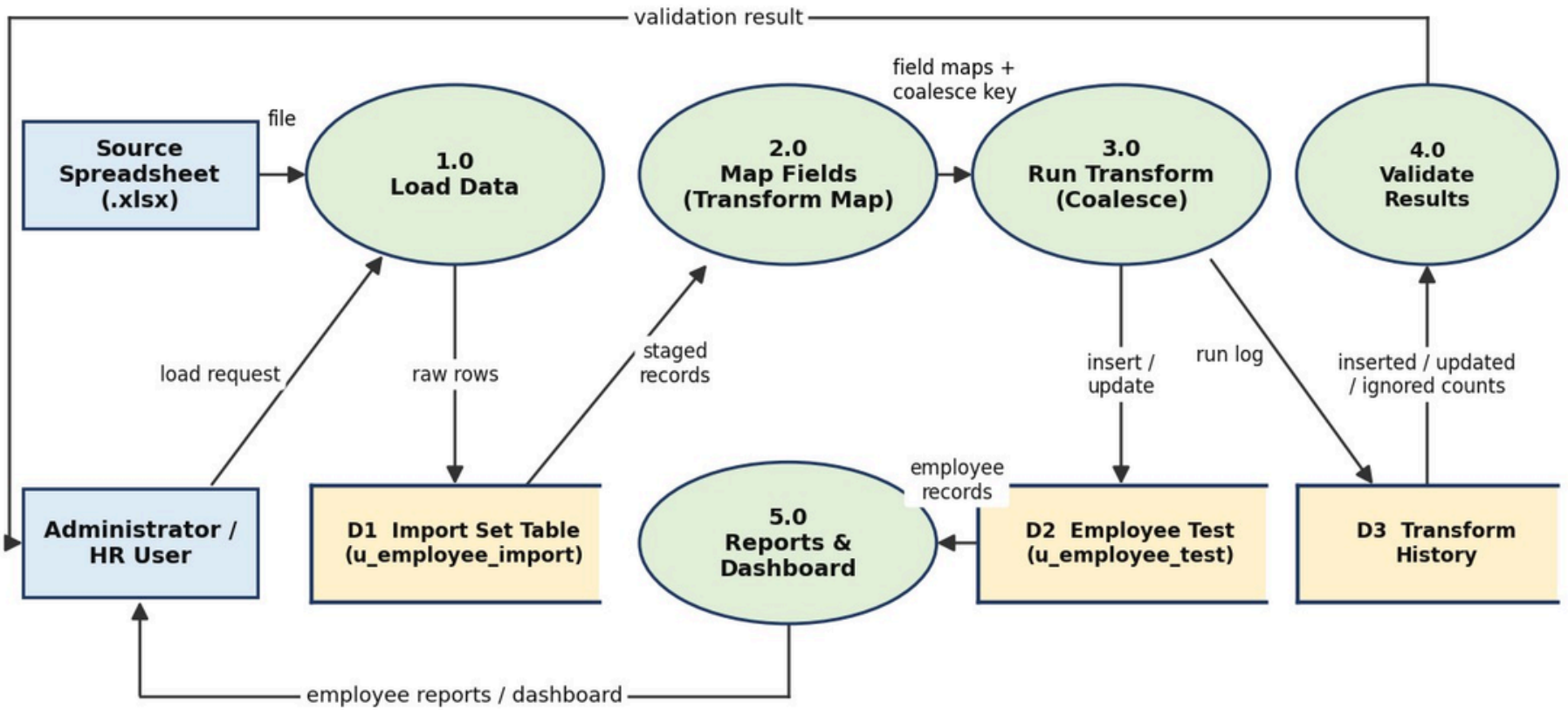
Date	30 September2026
Team ID	9100Spark
Project Name	Import Data using Transform Maps
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) shows how data enters and leaves the system, what changes the information, and where data is stored. The DFD below shows how employee data moves from the source spreadsheet through the ServiceNow Import Set table and Transform Map into the target Employee Test table, and how the results are validated and reported.

Element	Name	Description
External entity	Source Spreadsheet (.xlsx)	Sample Spreadsheet.xlsx with Employee ID, Name, Email, Department, Location
External entity	Administrator / HR User	Starts the import, checks the results and views reports and the dashboard
Process 1.0	Load Data	Loads the file through System Import Sets → Load Data into the Import Set table
Process 2.0	Map Fields (Transform Map)	Maps source fields to Employee Test fields; Employee ID is the Coalesce field
Process 3.0	Run Transform	Inserts new records and updates existing ones in the target table
Process 4.0	Validate Results	Checks the completion state, Transform History counts and the records in the target table
Process 5.0	Reports & Dashboard	Builds reports (department, location, list) and shows them on the Employee Analytics Dashboard
Data store D1	Import Set Table (u_employee_import)	Temporary staging table that holds the raw uploaded rows
Data store D2	Employee Test (u_employee_test)	Target table holding the final employee records
Data store D3	Transform History	Log of each transform run: total, inserted, updated, ignored, skipped and errors

Data Flow Diagram (Level 0/1):



Legend: rectangles = external entities; circles/ovals = processes; open-ended rectangles = data stores.

User Stories

Usethebelowtemplate to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story/Task	Acceptance criteria	Priority	Release
Administrator	Spreadsheet Preparation	USN-1	As an administrator, I can create a Google Spreadsheet with sample employee data (Employee ID, Name, Email, Department, Location).	The sheet contains all five columns with sample rows	High	Sprint-1
Administrator	Spreadsheet Preparation	USN-2	As an administrator, I can download the spreadsheet as Sample Spreadsheet.xlsx.	The .xlsx file is saved and ready for upload	High	Sprint-1
Administrator	Target Table Creation	USN-3	As an administrator, I can create the custom table Employee Test (u_employee_test) with String fields for the employee data.	All five fields appear on the Employee Test form	High	Sprint-1
Administrator	Data Loading	USN-4	As an administrator, I can load the .xlsx file using Load Data into a new Import Set table, Employee Import (u_employee_import).	A success message is shown and the rows are staged in the Import Set table	High	Sprint-2
Administrator	Field Mapping	USN-5	As an administrator, I can create a Transform Map (Sample Spreadsheet Import) from Employee Import to Employee Test and use Auto Map Matching Fields.	Field maps for all five fields are listed under the Transform Map	High	Sprint-2
Administrator	Transform & Validate	USN-6	As an administrator, I can run the Transform so the staged data is created in the target table.	The transform state shows Complete / Success and records appear in Employee Test	High	Sprint-3
Administrator	Transform & Validate	USN-7	As an administrator, I can validate the imported records and arrange the list columns using Personalize List Columns.	All imported records are visible with the correct field values	Medium	Sprint-3
Administrator	Duplicate Prevention	USN-8	As an administrator, I can enable Coalesce on the Employee ID field in the Transform Map.	The Employee ID field map shows Coalesce = true	High	Sprint-3
Administrator	Duplicate Prevention	USN-9	As an administrator, I can re-import a modified spreadsheet so existing employees are updated and only new ones are inserted.	Transform History shows the expected Inserts and Updates; re-importing the same file shows 0 inserts, 0 updates and all rows ignored	High	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	Reports & Dashboard	USN-10	As an administrator, I can enable Admin overrides on the pa_dashboards create ACL so dashboards can be created.	The PA_DASHBOARDS.FORM opens and a dashboard can be saved	Medium	Sprint-4
HR Manager	Reports & Dashboard	USN-11	As an HR manager, I can view reports for Employees by Department (pie chart), Employees by Location and the Employee List.	Each report runs and shows the current employee data	Medium	Sprint-4
HR Manager	Reports & Dashboard	USN-12	As an HR manager, I can see all three reports together on the Employee Analytics Dashboards dashboard.	The dashboard shows the three reports as widgets	Medium	Sprint-4
Administrator	Reports & Dashboard	USN-13	As an administrator, I can restrict who can view reports and the dashboard by users, groups and roles.	Only the selected users, groups or roles can open the shared reports	Low	Sprint-4